# Algorithms and Computability

Lecture 6: Nondeterministic Polynomial Time

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# Yesterday in Algorithms and Computability

We have compared algorithmic complexity:

- study of concrete algorithms and precise running times
- difference between $\mathcal{O}(n^2)$ and $\mathcal{O}(n^3)$ is huge
- running times depend on model of computation

and complexity theory:

- study of problems (languages) rather than algorithms
- difference between $\mathcal{O}(n^2)$ and $\mathcal{O}(n^3)$ may just depend on choice of model (but people still try to find optimal (for fixed model) algorithms)
- results should be valid for every model of computation

# Runtime of a Deterministic Turing Machine

## Definition

Let $M$ be a halting DTM.

- Let $time_M(w)$ denote the number of configurations in the unique run of $M$ on input $w$.
- Let $T\colon \mathbb{N} \to \mathbb{R}_{>0}$. We say that $M$ runs within time $T$ if $time_M(w) \leq T(|w|)$ for all inputs $w$.

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We are interested in classifying problems (languages) according to their asymptotic time-complexity.

## Definition

Let $T\colon \mathbb{N} \to \mathbb{R}_{>0}$ be a function. The complexity class $\textsc{Time}(T)$ is defined as

$$\textsc{Time}(T) = \{L(M) \mid M \text{ is a halting DTM that runs within time } \mathcal{O}(T)\}.$$

# The Complexity Class P

## Definition

The complexity class $\mathrm{P}$ (polynomial time) is defined as

$$\mathrm{P} = \bigcup_{k \geq 0} \textsc{Time}(n^k)$$

- Robust definition (can use other deterministic (!) models of computation, e.g., multi-tape Turing machines).
- Cobham's thesis: A problem can be efficiently computed if and only if it is in $\mathrm{P}$.

# Agenda

1. **Motivation**
2. Nondeterministic Time Complexity
3. Polynomial-time Reductions

# Knapsack

Recall the Knapsack problem:

> *A thief has a knapsack holding at most $W$ pounds of loot, robs a store that has items $1, \ldots, n$ of weight $w_j$ and value $c_j$. What is the maximal value the thief can put in his knapsack?*

# Knapsack

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As a decision problem:

$$\{W, T, w_1, \ldots, w_n, c_1, \ldots, c_n \in \mathbb{N} \mid \exists b_1, \ldots, b_n \in \{0, 1\} \text{ s.t. } \sum_{j=1}^{n} b_j \cdot w_j \leq W \text{ and } \sum_{j=1}^{n} b_j \cdot c_j \geq T\}$$

# Knapsack

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Consider a backpack of size $W = 100$, the threshold $T = 200$, and the following items:

| item $j$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| weight $w_j$ | 10 | 20 | 60 | 25 | 50 | 35 |
| value $c_j$ | 5 | 70 | 20 | 105 | 30 | 35 |

Is $b_1 = 0$, $b_2 = 1$, $b_3 = 0$, $b_4 = 1$, $b_5 = 1$, $b_6 = 0$ a valid solution?

# Knapsack

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Is $b_1 = 0$, $b_2 = 1$, $b_3 = 0$, $b_4 = 1$, $b_5 = 1$, $b_6 = 0$ a valid solution? Yes, because

$$w_2 + w_4 + w_5 = 20 + 25 + 50 = 95 \leq 100 = W$$

and

$$c_2 + c_4 + c_5 = 70 + 105 + 30 = 205 \geq 200 = T$$

# Knapsack

$$\{W, T, w_1, \ldots, w_n, c_1, \ldots, c_n \in \mathbb{N} \mid \exists b_1, \ldots, b_n \in \{0,1\} \text{ s.t. } \sum_{j=1}^{n} b_j \cdot w_j \leq W \text{ and } \sum_{j=1}^{n} b_j \cdot c_j \geq T\}$$

Now, consider a backpack of size 15, the threshold $T = 50$, and the following items:

| item $j$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| weight $w_j$ | 2 | 3 | 5 | 7 | 1 | 4 | 1 |
| value $c_j$ | 10 | 5 | 15 | 7 | 6 | 18 | 3 |

Does it have a valid solution?

# Satisfiability

$$\varphi = (x_0 \vee x_2) \wedge (x_0 \vee \neg x_3) \wedge (x_1 \vee \neg x_3) \wedge (x_1 \vee \neg x_4) \wedge (x_2 \vee \neg x_4) \wedge$$
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Is $x_0 = x_1 = x_2 = x_3 = x_4 = x_5 = x_6 = 1$ a satisfying assignment?

# Satisfiability

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# Satisfiability

$\varphi = 1$ **Yes**!

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# Satisfiability

$\varphi = 1$ **Yes!**

Does

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have a satisfying assignment?

# Similarities

- Knapsack:

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- Boolean Satisfiability:

$$\mathrm{SAT} = \{\varphi \mid \exists \text{ satisfying assignment for } \varphi\}$$

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- Knapsack:

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## Similarities

- Both problems ask for the existence of a certificate "proving" that the input is in the language, i.e., a vector $b_1 \cdots b_n$ or an assignment.
- Certificates are easy to verify, i.e, in polynomial time.
- Certificates are short, i.e., at most polynomial in the input length.

# More

Many other important problems share these traits:

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$$\{G \mid G \text{ is a graph with a hamiltonian path}\}$$

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# Nondeterminism to the Rescue

Nondeterminism allows a Turing machine to find a certificate: For example, to turn any sequence of $n$ consecutive $X$'s into all 0/1 vectors of length $n$ use the following transitions:

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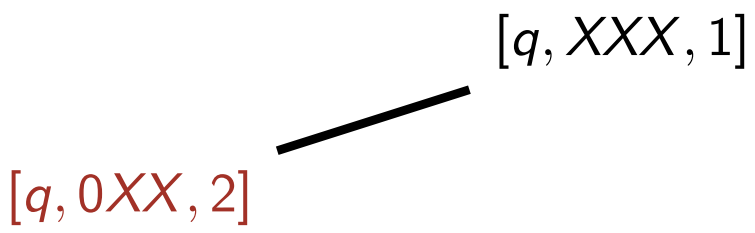

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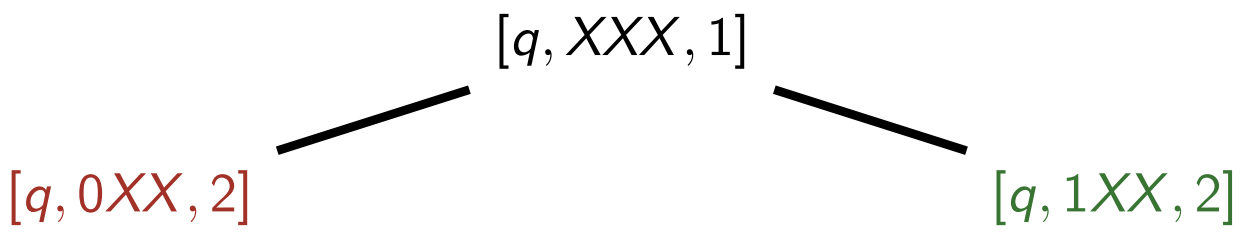

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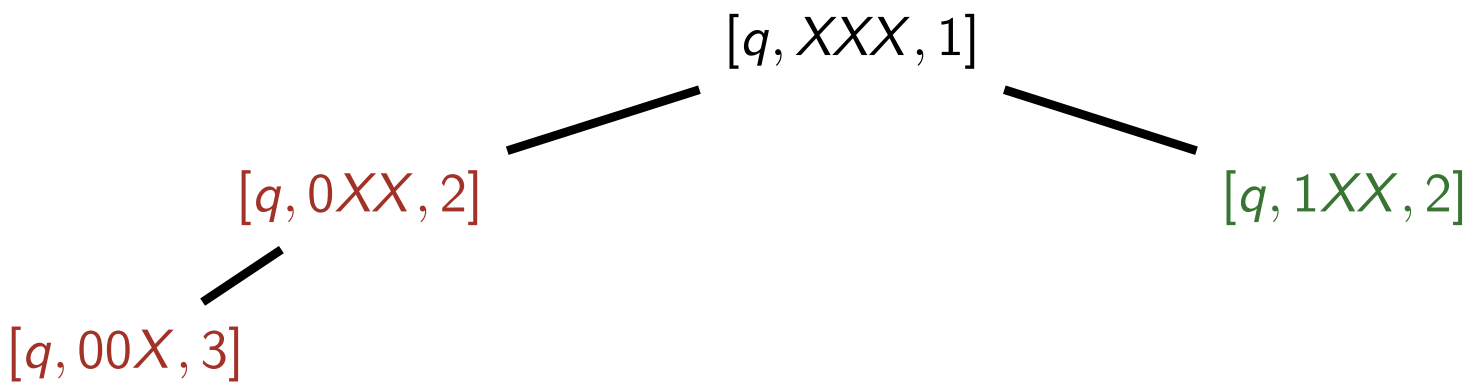

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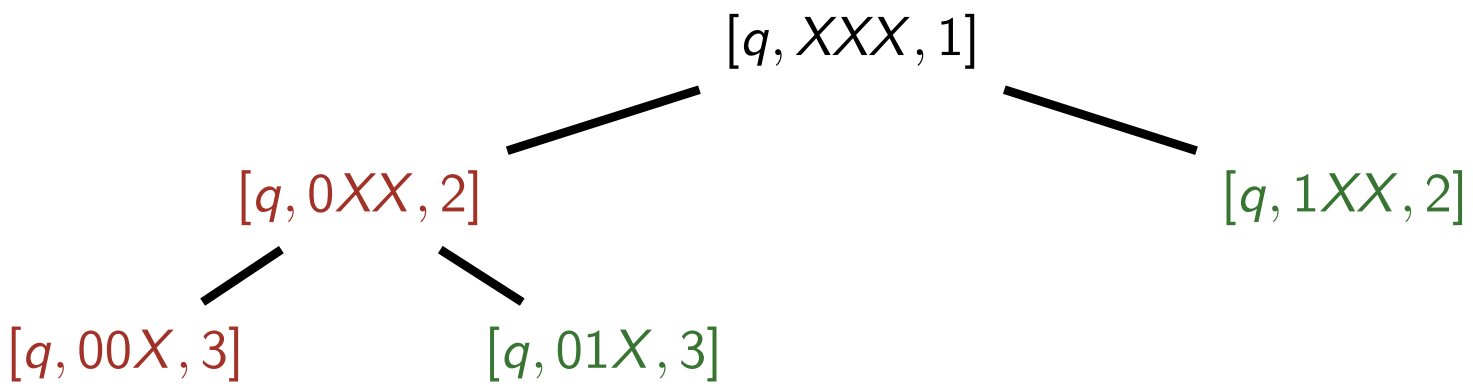

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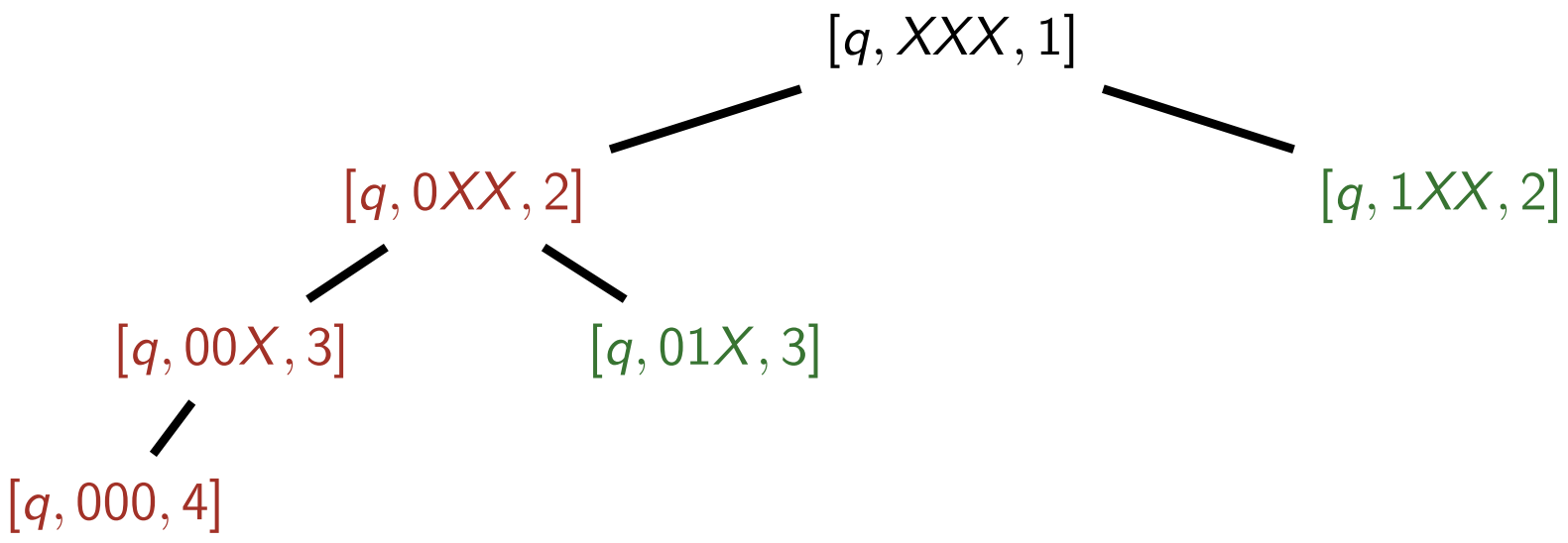

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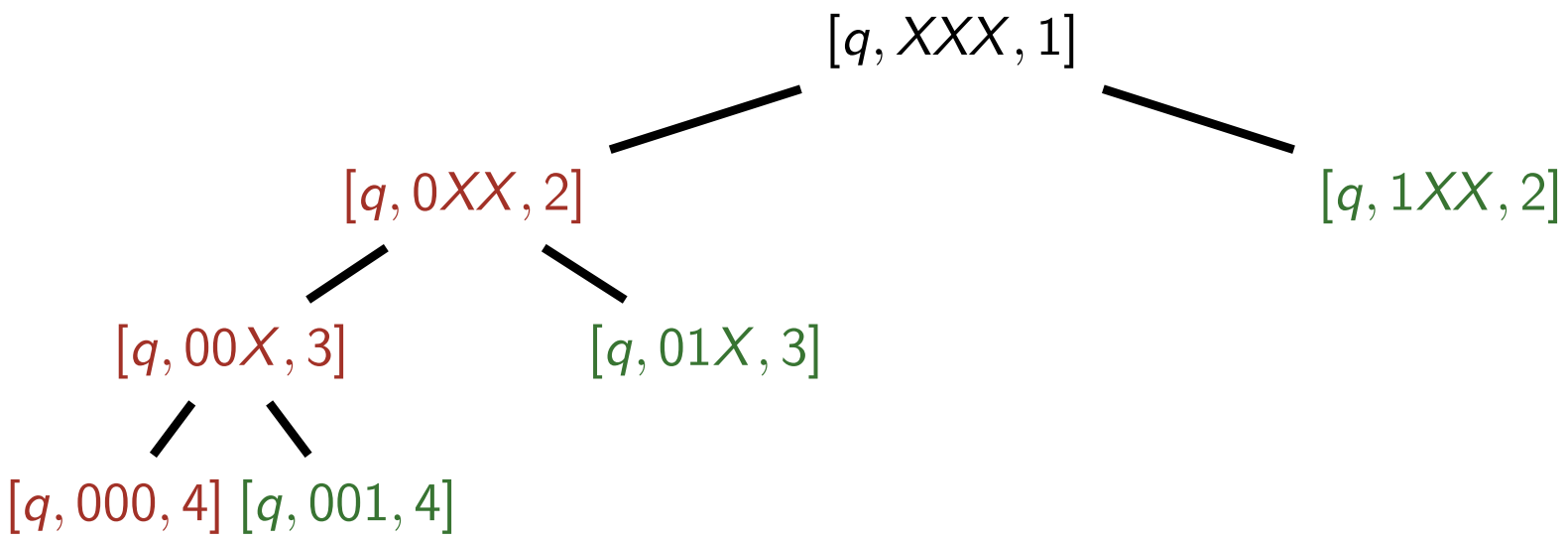

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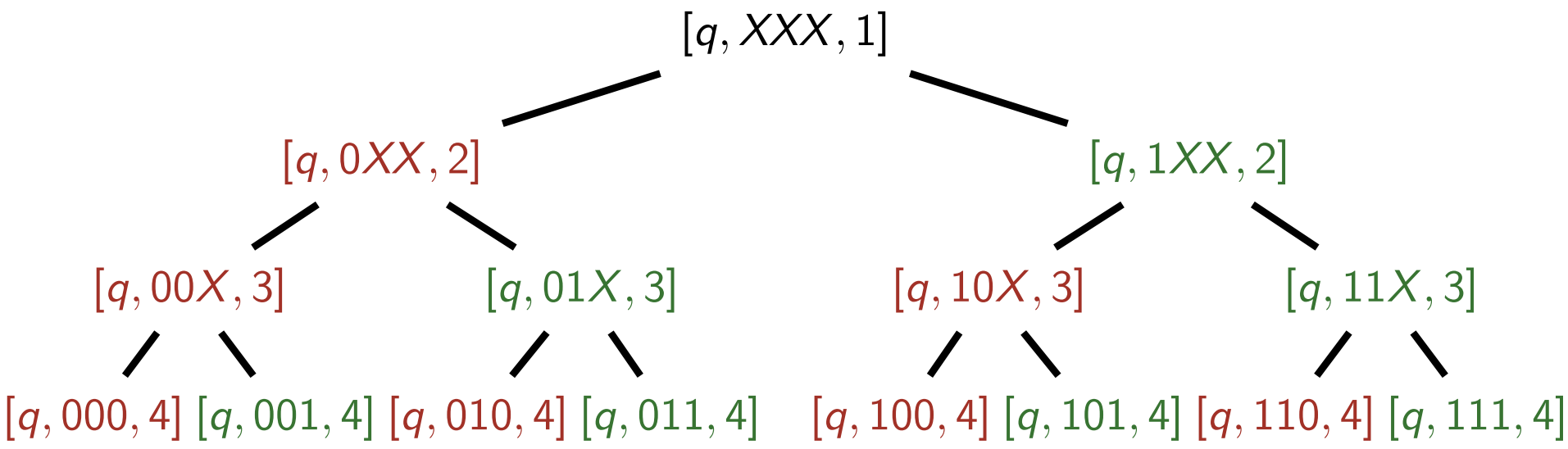

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# Agenda

1. Motivation
2. **Nondeterministic Time Complexity**
3. Polynomial-time Reductions

# Runtime of a Nondeterministic Turing Machine

## Definition

Let $M$ be a halting NTM.

- Let $time_M(w)$ denote the maximal number of configurations in a branch of the computation tree of $M$ on input $w$.
- Let $T\colon \mathbb{N} \to \mathbb{R}_{>0}$. We say that $M$ runs within time $T$ if $time_M(w) \leq T(|w|)$ for all inputs $w$.

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We are still interested in classifying problems (languages) according to their asymptotic time-complexity.

## Definition

Let $T\colon \mathbb{N} \to \mathbb{R}_{>0}$ be a function. The complexity class $\mathrm{NTIME}(T)$ is defined as

$$\mathrm{NTIME}(T) = \{L(M) \mid M \text{ is a halting NTM that runs within time } \mathcal{O}(T)\}.$$

# The Complexity Class NP

## Definition

The complexity class $\mathrm{NP}$ (polynomial time) is defined as

$$\mathrm{NP} = \bigcup_{k \geq 0} \mathrm{NTime}(n^k)$$

- Robust definition (can use other nondeterministic models of computation, e.g., multi-tape Turing machines).
- Contains $\mathrm{P}$, as every halting DTM with polynomial time complexity is a halting NTM with polynomial time complexity:

## Remark

$\mathrm{P} \subseteq \mathrm{NP}$.

# Another Example

- A nondeterministic polynomial-time algorithm for Clique.
- Recall that we encode a graph $G$ by (a word over $\mathbb{B}$ representing) its adjacency matrix, i.e., entry $(i, j)$ is one if and only if there is an edge from the $i$-th vertex to the $j$-th one.

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In fact, the other problems we have seen are also in $\mathrm{NP}$:

- Nonprimality
- Hamiltonian path
- 3-coloring
- and many more.

# Agenda

1. Motivation
2. Nondeterministic Time Complexity
3. **Polynomial-time Reductions**

# Polynomial-time Computable Functions

## Definition

A function $f : \Sigma^* \to \Gamma^*$ is a polynomial-time computable function if and only if there exists a DTM $M_f$ with polynomial time-complexity that on every input $w \in \Sigma^*$

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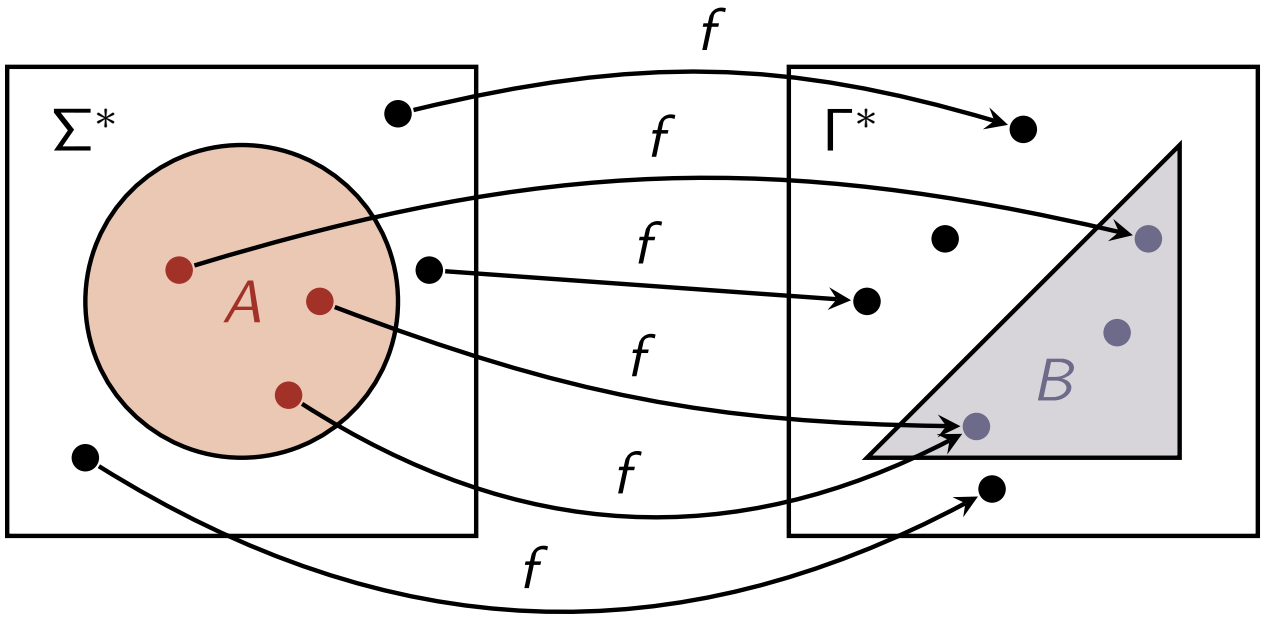
## Examples

- $f(w) = ww$ is polynomial-time computable.
- $f(\ulcorner m \urcorner \# \ulcorner n \urcorner) = \ulcorner m \cdot n \urcorner$ is polynomial-time computable (where $\ulcorner n \urcorner$ denotes the binary encoding of $n \in \mathbb{N}$).

# Polynomial-time Reduction




## Definition

Let $A \subseteq \Sigma^*$ and $B \subseteq \Gamma^*$ be languages. We say that $A$ is polynomial-time reducible to $B$, written $A \leq_P B$, if and only if

1. there is a polynomial-time computable function $f : \Sigma^* \to \Gamma^*$ such that
2. for every $w \in \Sigma^*$: $w \in A \Leftrightarrow f(w) \in B$.

# Applications

Recall: Let $A \leq_m B$.

- If $B$ is computable, then $A$ is computable.
- If $B$ is computably-enumerable, then $A$ is computably-enumerable.

A similar result holds for polynomial-time reductions and the complexity classes $\mathrm{P}$ and $\mathrm{NP}$.

**Theorem**

*Let $A \leq_P B$.*

1. *If $B \in \mathrm{P}$, then $A \in \mathrm{P}$.*
2. *If $B \in \mathrm{NP}$, then $A \in \mathrm{NP}$.*

# Proof

We prove "If $B \in \mathrm{P}$, then $A \in \mathrm{P}$". The proof for $\mathrm{NP}$ is analogous.

- Let $M$ be a halting DTM for $B$ that runs within time $\mathcal{O}(n^k)$.

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- $M''$ runs within time $(n^{k'})^k = n^{k' \cdot k}$.
- Hence, $A \in P$.

# Conclusion

We have seen:

- the complexity class $\mathrm{NP}$: The class of languages accepted by polynomial-time NTM's.
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Recall: $\mathrm{P} \subseteq \mathrm{NP}$.

**The (literally) Million Dollar Question**

Is verifying certificates easier than finding certificates or not: Is $\mathrm{P} = \mathrm{NP}$ or $\mathrm{P} \subsetneq \mathrm{NP}$?

- One of the deepest, most challenging, and most important questions of (theoretical) computer science.
- More on that next week.

# Reading

Sections 3.1 to 3.4 of "Computability and Complexity" (pages 141 to 178):

## Note

- The book uses slightly different notation and definitions.
- These sections also cover the complement class of $\mathrm{NP}$ (called $\mathrm{coNP}$). You can read about it, but this topic is not exam-relevant.